

CyberQuantum Iberia SL

Tender: CTTI-2026-36

Subministrament, instal·lacio i posada en produccio d'equipaments de distribucio de clau quantica

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CyberQuantum Iberia SL submits this technical proposal for CTTI tender CTTI-2026-36. Our offer combines robust deployment risk management, a mature monitoring platform, and standards-compliant QKD equipment to deliver a secure and operationally sound quantum key distribution service.

1.1 PLA DE MIGRACIO — PROJECT TEAM

Our Programme Manager brings 5 years of telecommunications project management experience in optical transport and network service delivery. While his experience base is in classical telecoms rather than quantum-specific deployments, he has undergone internal product training on QKD system architecture. Two CCNP-certified Network Administrators are assigned. Neither profile references prior experience with QKD communication systems specifically. Level 3 escalation support will be provided by senior technical resources within CyberQuantum's engineering organisation; specific certifications and manufacturer accreditation status for these resources are not detailed in this proposal.

1.2 EXECUCIO CRITICA I DESPLEGAMENT

Team for Execution Phase

The solution architect assigned has 5 years of network architecture experience and describes his working knowledge of QKD systems as familiarity with QKD concepts, gained through internal technical education and vendor briefings. He has not held a prior architect role on a live QKD deployment project and does not hold CCDP or NSE 7 certification.

Risk Identification

Risk 1 — Optical signal degradation at legacy fibre nodes: Probability high. Impact: service commissioning delay or reduced SKR. Mitigation: mandatory pre-installation OTDR survey and physical fibre qualification at every node prior to equipment delivery. Formal qualification acceptance criterion defined.

Risk 2 — KMS-QKD synchronisation latency under high concurrent load: Probability medium-high. Impact: service degradation for latency-sensitive key consumers. Mitigation: dedicated management VLAN provisioned for all QKD and KMS control plane traffic with QoS policies enforcing strict priority queuing for key distribution flows.

Risk 3 — Firmware incompatibility between QKD emitter and KMS versions during phased rollout: Probability medium. Impact: node activation failures. Mitigation: staged firmware validation protocol at each node prior to production commissioning, including verified rollback procedure. Compatibility matrix maintained and updated after every firmware change.

Service Monitoring Tool

Comprehensive NOC monitoring platform covering the full CTTI QKD estate. Tracks Secure Key Rate in real time with automated alerting at configurable SKR threshold breaches. Web-based dashboard accessible to both CyberQuantum operations staff and authorised CTTI personnel. Data refreshed on a 3-minute cycle, exceeding the standard 5-minute frequency. Native SNMP integration configured for CTTI infrastructure from day one.

Documentation

Full technical documentation will be delivered in Catalan for all supplied equipment and systems. Handover plan will specify team composition, training sessions and knowledge transfer milestones.

Equipment — Flexibility

CV-QKD system meets PPT minimum performance thresholds: net symmetric SKR ≥ 1 kbps per pair at 10 dB. Equipment ships in standard 19-inch rack format within the 2U height constraint. Emitter and receiver units can be replaced independently without requiring full system renewal.

Equipment — Technical Specifications

KMS component is FIPS 140-2 certified and compliant with ETSI GS QKD 004 and 014. Power supply redundancy is provided on all primary units.

Equipment — Robustness and Resilience

Power supply redundancy provided on all primary units. No additional component-level redundancy specifications are detailed in this submission.

1.3 ANALISI DE LES DADES — DATA ANALYSIS TEAM

For the data analysis phase, we will assign one analyst who will work across the measurement and reporting deliverables defined in PPT Section 5.4. No specific quantum traffic analysis background is cited for this profile.

1.4 PLA DE DEVOLUCIO DEL SERVEI

The handover plan commits to two structured knowledge transfer sessions meeting the PPT minimum. Sessions will cover system architecture and operational procedures. All documentation delivered in Catalan. A 6-month post-handover support period is included in the base proposal. Transition team composition is not specified in this document.

ADVISORY: This document is a synthetic technical proposal generated for evaluation agent testing purposes only. All company names, personnel, certifications, and project references are fictional.